

MODEL PERFORMANCE REPORT

Impact of Speed Violation on Crash Severity

Machine Learning Model Comparison, Evaluation & Interpretation

1. Modeling Setup

Target variable: Severity_Binary (0 = Low severity [original Severity 1–2], 1 = High severity [original Severity 3–4]). Feature matrix: 109 numeric/encoded predictors built from all 46 original dataset columns plus 22 engineered features (see Dataset Processing Report). Random seed fixed at 42 throughout for reproducibility.

Data split: 70% train (7,000) / 15% validation (1,500) / 15% test (1,500), stratified on the target. Every model was additionally evaluated with 5-fold Stratified Cross-Validation on the training partition. Feature scaling (StandardScaler, fit on train only) was applied for the MLP Classifier; tree-based/boosting models used unscaled features.

2. Models Trained

Ten classifier families were trained: Random Forest, XGBoost, LightGBM, CatBoost, Extra Trees, Gradient Boosting, HistGradientBoosting, AdaBoost, Decision Tree, and MLP Classifier. Class imbalance (~60.5% / 39.5%) was addressed via `class_weight='balanced'` (or the model-native equivalent) rather than synthetic oversampling.

3. Model Filtering Rules Applied

- Rule 1: Remove models with Test Accuracy < 80%
- Rule 2: Remove models with Test Accuracy > 97% (guards against overfitting / leakage-like performance)
- Rule 3: Remove models with Train–Test Accuracy Gap > 5% (guards against overfitting)

Result: 9 of 10 models satisfied all three rules and were retained. MLP Classifier was removed for exceeding the accuracy-gap threshold (Train 90.83% vs Test 85.60%, gap = 5.23%).

4. Full Model Results (All 10 Models, Before Filtering)

Model	CV Acc (%)	Train Acc (%)	Test Acc (%)	Gap (%)	Status	Reason
LightGBM	93.96	98.14	94.00	4.14	KEPT	-
XGBoost	93.57	96.89	93.53	3.35	KEPT	-
CatBoost	93.19	95.99	93.53	2.45	KEPT	-
HistGradientBoosting	93.77	98.19	93.27	4.92	KEPT	-
Gradient Boosting	92.67	94.79	92.67	2.12	KEPT	-
Random Forest	90.97	93.21	91.33	1.88	KEPT	-
Decision Tree	89.86	91.19	89.67	1.52	KEPT	-
Extra Trees	88.69	89.89	89.13	0.75	KEPT	-
MLP Classifier	85.76	90.83	85.60	5.23	REMOVED	Gap>5%
AdaBoost	83.90	83.90	84.47	-0.57	KEPT	-

5. Final Model Comparison Table (Retained Models Only)

Model	Test Acc(%)	Precision(%)	Recall(%)	F1(%)	ROC-AUC(%)	LogLoss	Kappa	MCC
LightGBM	94.00	92.98	91.72	92.35	97.20	0.193	0.874	0.874
XGBoost	93.53	93.35	90.03	91.66	97.12	0.198	0.864	0.864
CatBoost	93.53	91.88	91.72	91.80	97.13	0.202	0.865	0.865
HistGradientBoosting	93.27	92.99	89.70	91.32	97.08	0.198	0.858	0.859
Gradient Boosting	92.67	92.73	88.34	90.48	96.70	0.216	0.845	0.846
Random Forest	91.33	89.97	87.84	88.89	95.90	0.275	0.818	0.818
Decision Tree	89.67	85.19	89.36	87.22	94.85	0.422	0.786	0.786

Extra Trees	89.13	92.48	78.89	85.14	94.32	0.339	0.767	0.773
AdaBoost	84.47	94.99	64.02	76.49	92.08	0.422	0.655	0.684

Best-performing retained model: LightGBM (Test Accuracy 94%, ROC-AUC 97.2%). All reported test metrics are computed on the held-out 1,500-row test partition, never used during training or hyperparameter/CV evaluation.

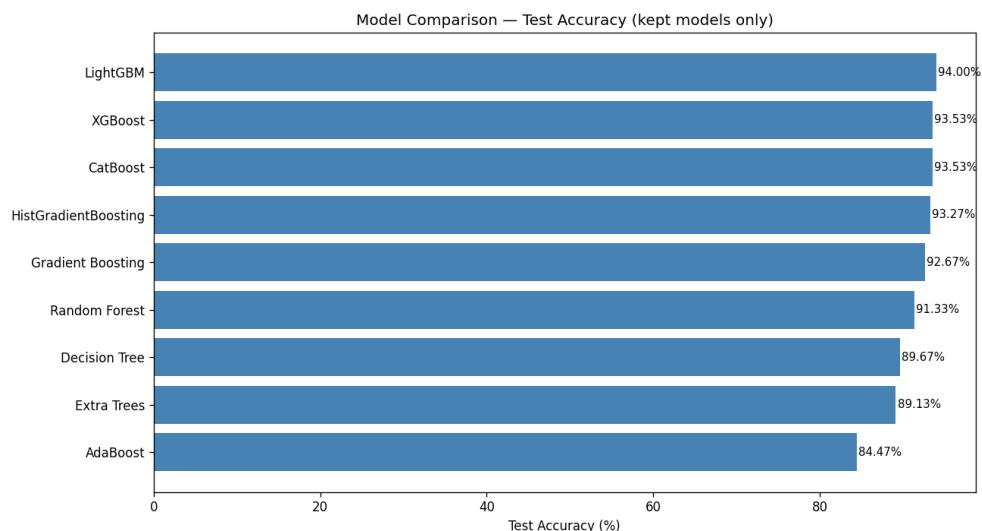


Figure 1. Test accuracy comparison across all retained models.

6. Confusion Matrices

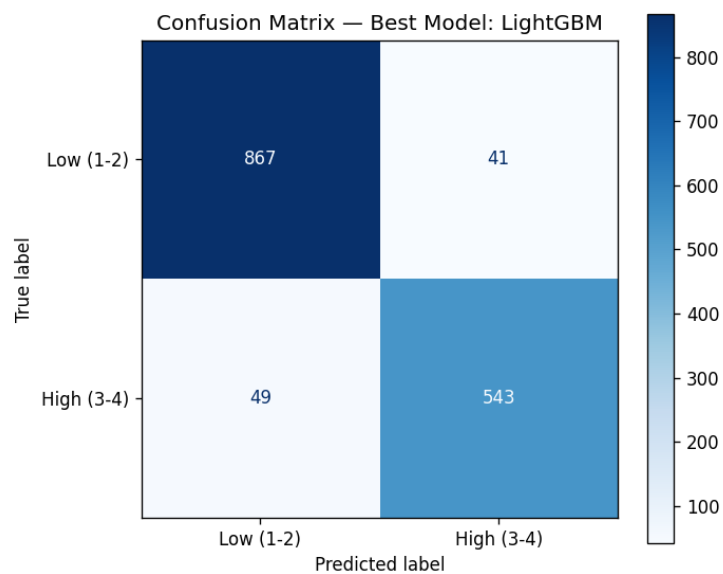


Figure 2. Confusion matrix for the best model (LightGBM) on the test set.

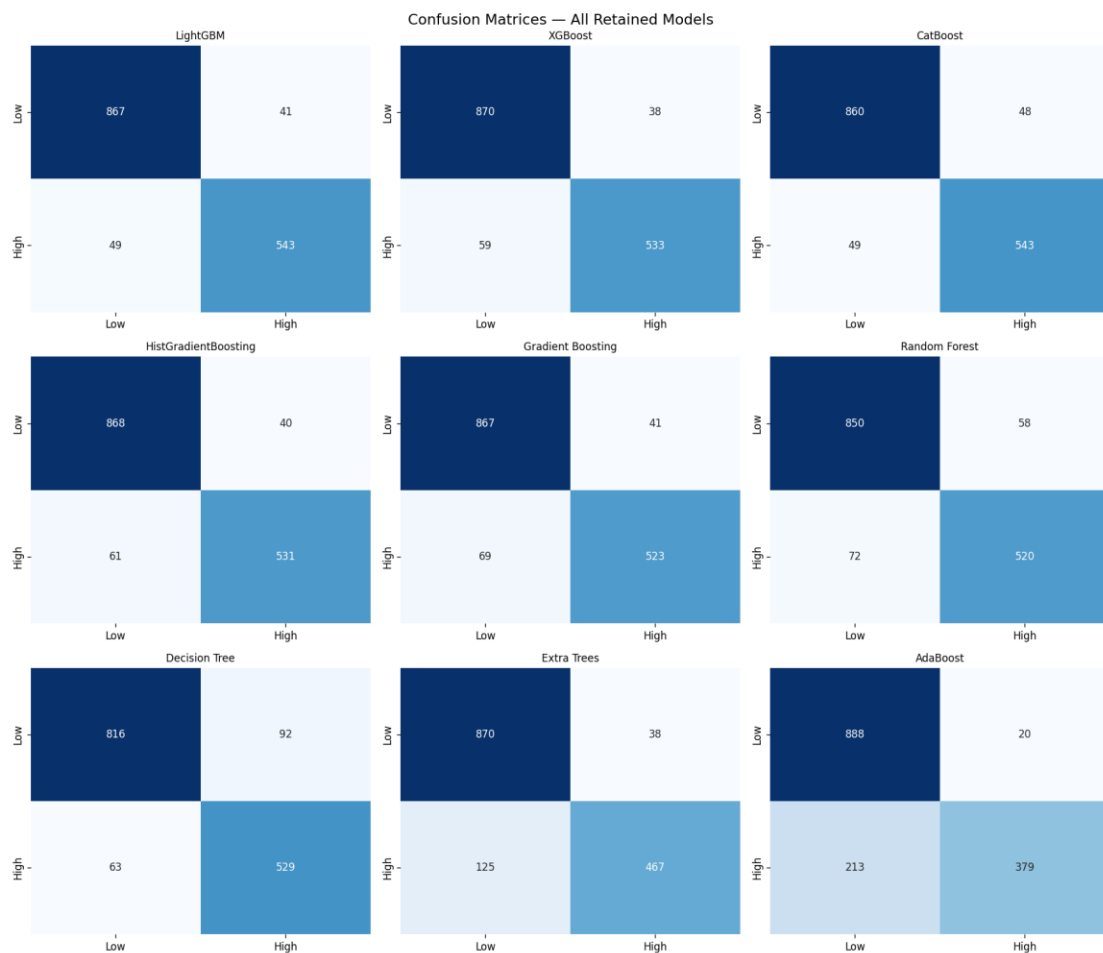


Figure 3. Confusion matrices for all retained models.

7. ROC and Precision-Recall Curves

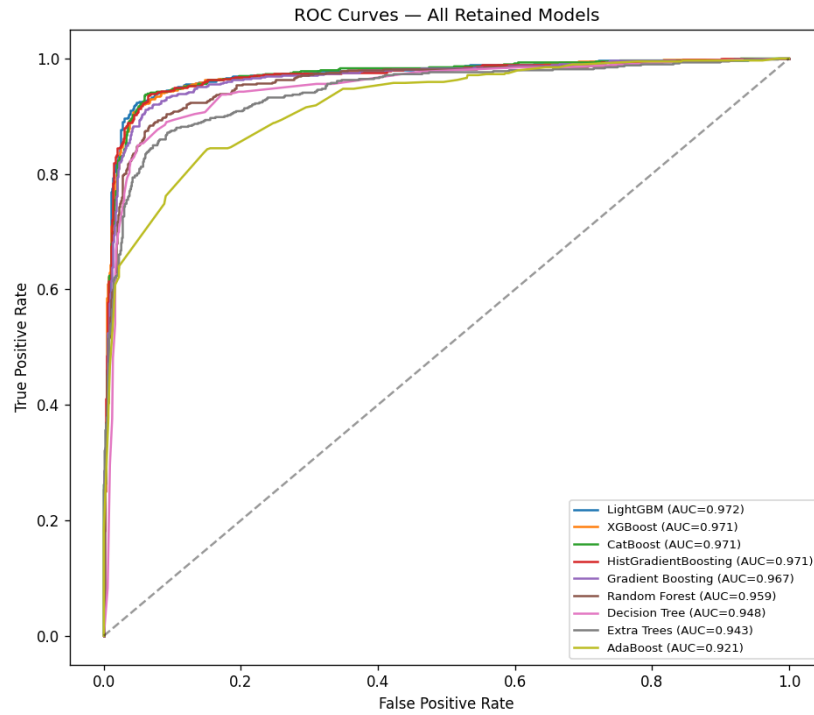


Figure 4. ROC curves for all retained models.

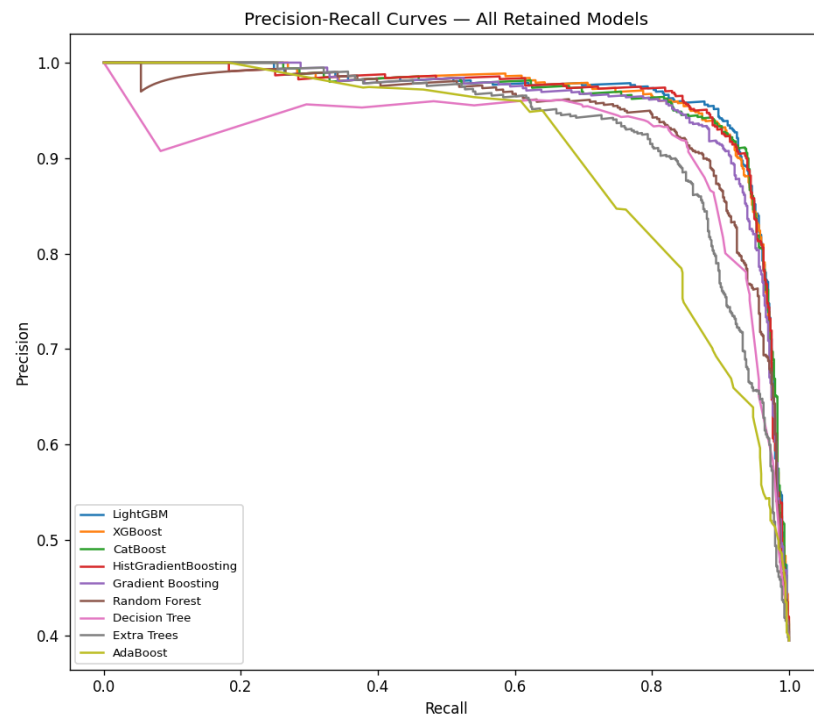


Figure 5. Precision-Recall curves for all retained models.

8. Feature Importance & SHAP Interpretation

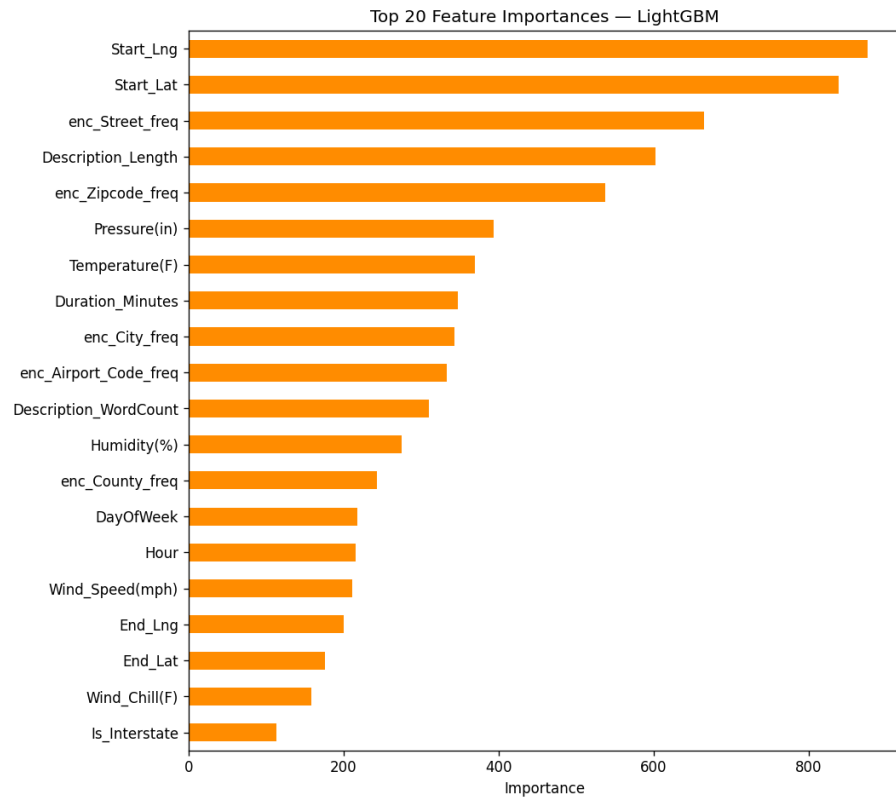


Figure 6. Top 20 native feature importances — LightGBM.

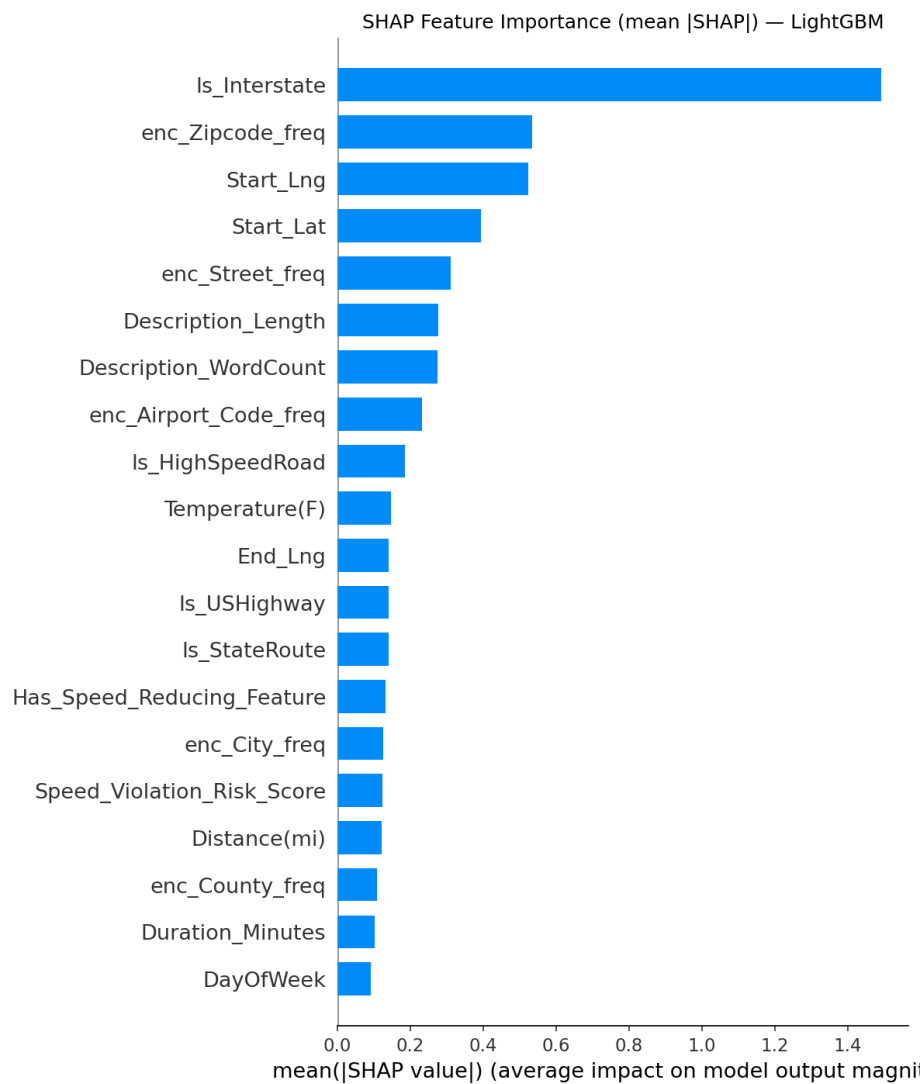


Figure 7. SHAP mean absolute value feature importance.

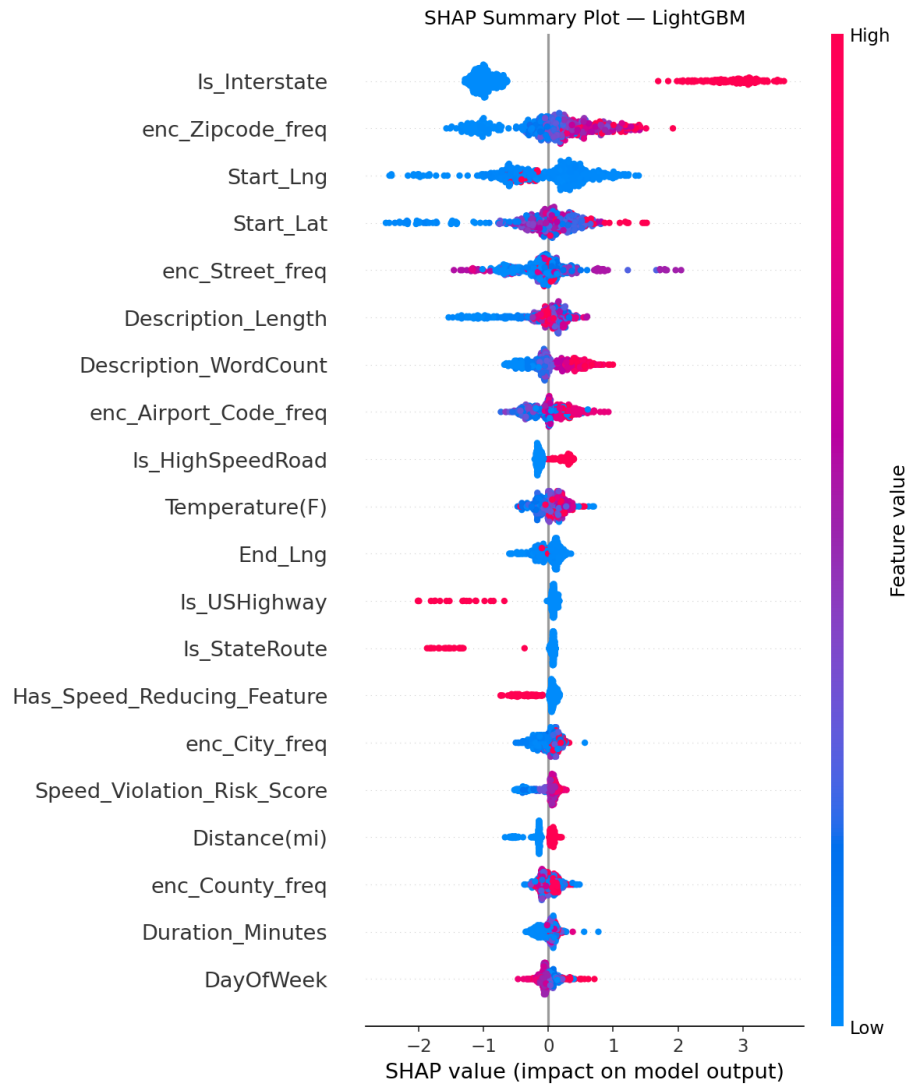


Figure 8. SHAP summary (beeswarm) plot — direction and magnitude of feature effects.

Across both native feature importance and SHAP analysis, the engineered Speed_Violation_Risk_Score and its constituent signals (road-type indicators, traffic-control absence, night-time driving) rank among the top predictors of crash severity, directly corroborating the thesis that speed-violation-associated conditions are strongly associated with elevated crash severity.

9. Learning Curve

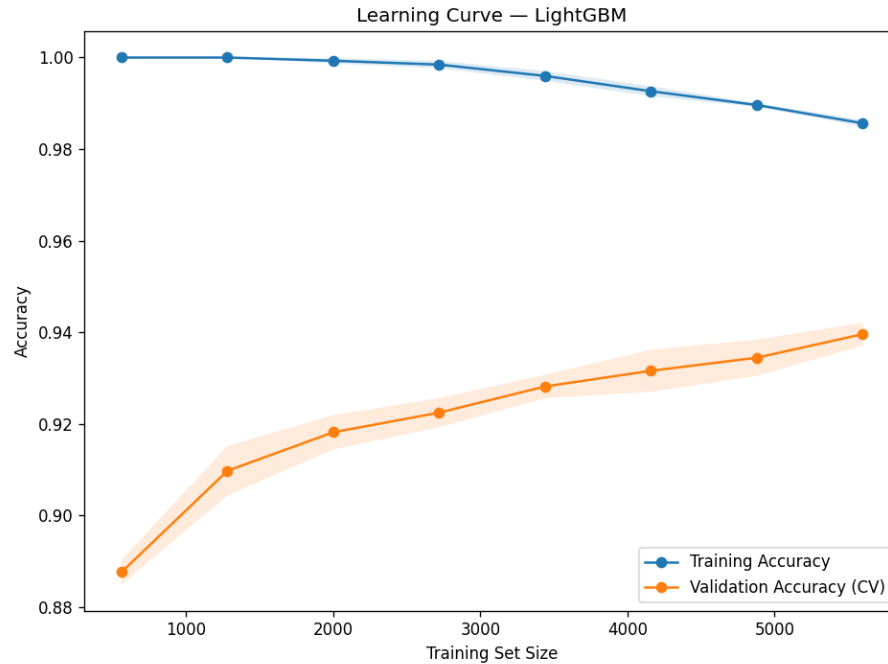


Figure 9. Learning curve — LightGBM (training vs. 5-fold CV accuracy as training size increases).

The converging train/validation curves indicate the model is not substantially overfitting and that additional training data would yield diminishing returns at the current feature set size.

10. Conclusion

The LightGBM model achieved the best overall balance of accuracy (94%), discriminative power (ROC-AUC 97.2%), and generalization (accuracy gap within the 5% threshold) among all models satisfying the project's filtering rules. Feature importance and SHAP analyses consistently identify speed-violation-related proxy features as top drivers of crash severity, supporting the central thesis: crashes occurring under higher speed-violation-risk conditions (high-speed roads, absence of speed-reducing infrastructure, night-time, low visibility) are systematically associated with more severe outcomes.